

Introduction to ETL Process



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Technical Trainers



SoftUni



Software University

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1. Introduction to ETL and ELT

- What is ETL?
- ETL Architecture
- What is ELT?
- ETL vs ELT

2. Demonstration of Simple ETL

- Scenario Sales Data ETL
- Steps

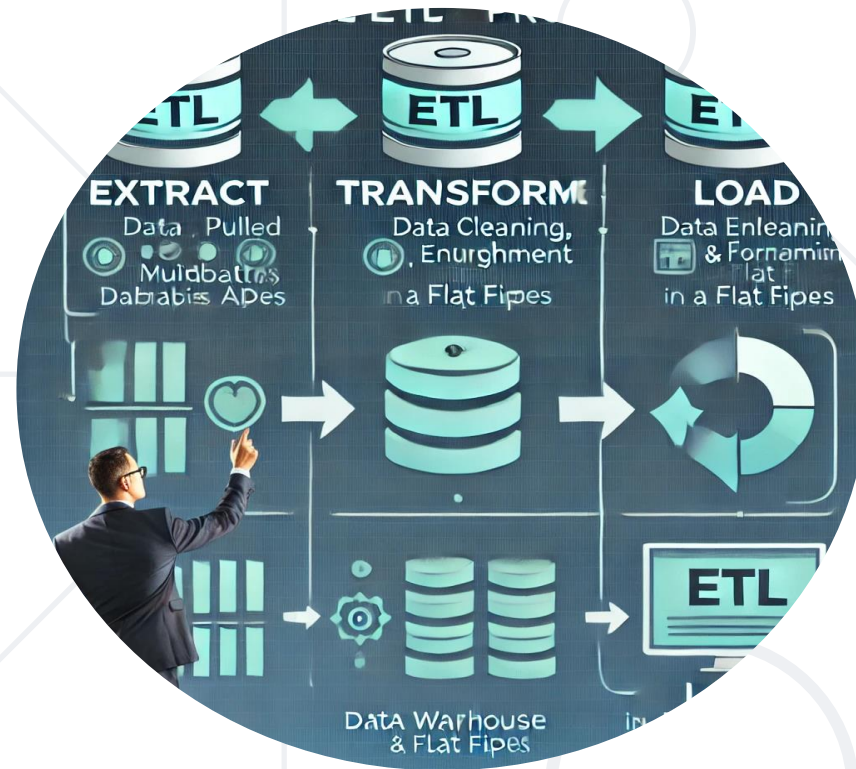


3. Modern ETL & Data Transformation Tools

- Data Transformation Tools
- Apache Airflow
- DBT (Data Build Tool)
- Other Tools

4. Use Cases



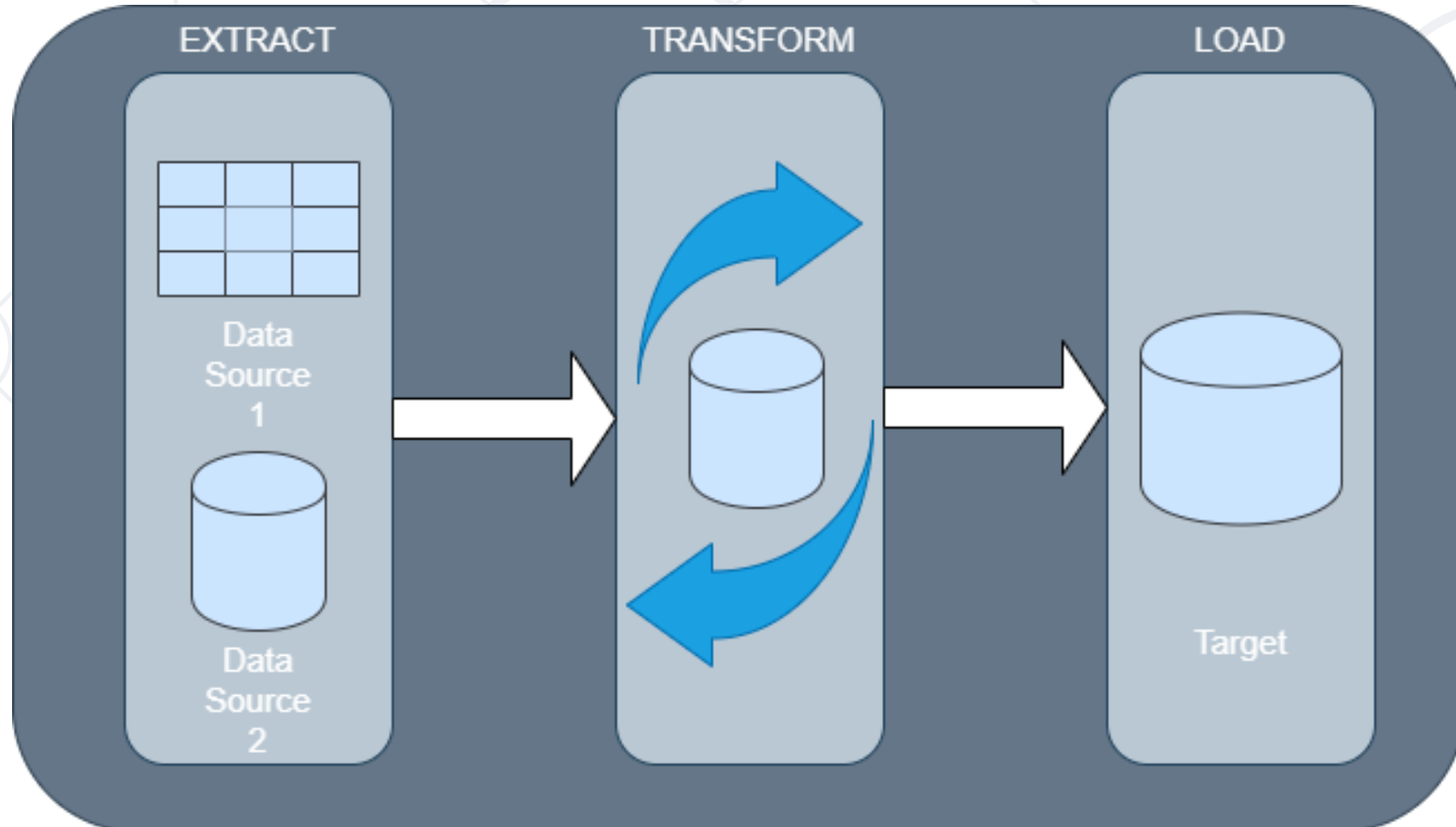


ELT and ETL

Introduction to ETL, Demonstration and Tools

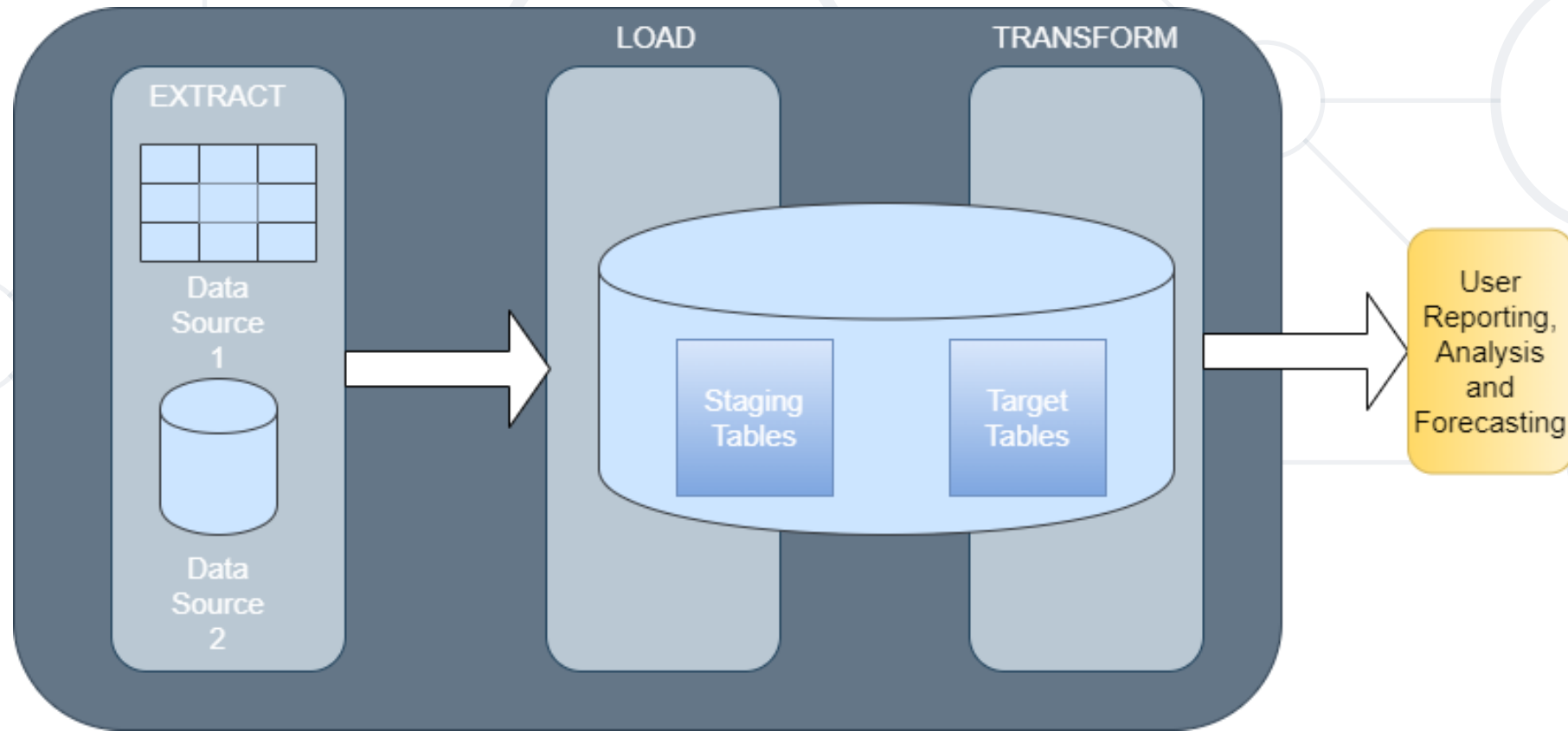
What is ETL?

- What is **ETL** – **E**xtract **T**ransform **L**oad is a **data integration process**
 - **Extract** – Data is gathered from multiple sources
 - **Transform** – Data is cleaned, formatted and enriched to meet business requirements
 - **Load** – Processed data is stored in a target system (e.g. Data Warehouse, Data Lake, Database, APIs)



- What is **ELT** – **E**xtract **L**oad **T**ransform is a **modern data integration** approach designed for cloud-based data processing
 - **Extract** – Data is pulled from various source systems
 - **Load** – Raw unprocessed data is loaded directly into the target systems
 - **Transform** – Data is cleaned, formatted and enriched to meet business requirements

ELT Architecture



Aspect	ETL (Extract, Transform, Load)	ELT (Extract, Load, Transform)
Process Order	Data is transformed before loading into the target system	Data is loaded before transforming in the target system
Transformation Location	Performed for external ETL tools or servers	Performed within the target system (e.g. Data Warehouse)
Data Stored	Transformed, clean data is stored	Raw, unprocessed data is stored, allowing for flexibility
Performance	Slower for large datasets as transformations occur before loading	Faster for big data due to direct loading and cloud-based transformations

- Demonstration of **Simple ETL Pipeline**
 - Step 1 – **Extract raw data from source (CSV)**

```
Import pandas as pd
raw_data = pd.read_csv("sales_data.csv")
print("Extracted Data")
print(raw_data.head())
```

- Step 2 – **Clean and process raw data** (e.g. handle missing values, calculate new fields)

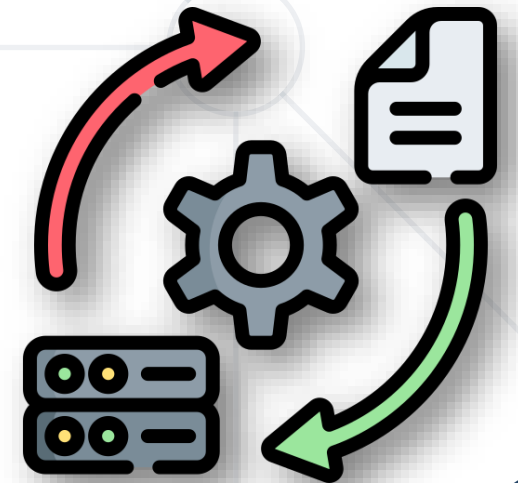
```
raw_data["sales_amount"] = raw_data["sales_amount"].fillna(0)
raw_data["total_revenue"] = raw_data["price"] *
raw_data["quantity"]
cleaned_data = raw_data[raw_data["total_revenue"] > 0]
print("Transformed Data")
print(cleaned_data.head())
```

- Step 3 – Load the transformed data into source (PostgreSQL database)

```
import psycopg2
connection = psycopg2.connect(host = "localhost", database =
"sales_db",
    user = "postgres",
    password = "password"
)
cursor = connection.cursor()
cursor.execute("""CREATE TABLE IF NOT EXIST
sales_summary(product_id INT, quantity = INT, total_revenue
NUMERIC)""")
```

```
for index, row in cleaned_data.iterrows():
    cursor.execute("""
        INSERT INTO sales_summary (product_id, quantity,
                                   total_revenue) VALUES (%s, %s, %s)
    """, (row["product_id"], row["quantity"],
row["total_revenue"]))
    connection.commit()
    cursor.close()
    connection.close()
    print("Data loaded into the database successfully")
```

- **Data Transformation Tools**
 - Data Extraction with Pandas or Request
 - Transformation Using Pandas or SQL Scripts
 - Loading Data into Target System



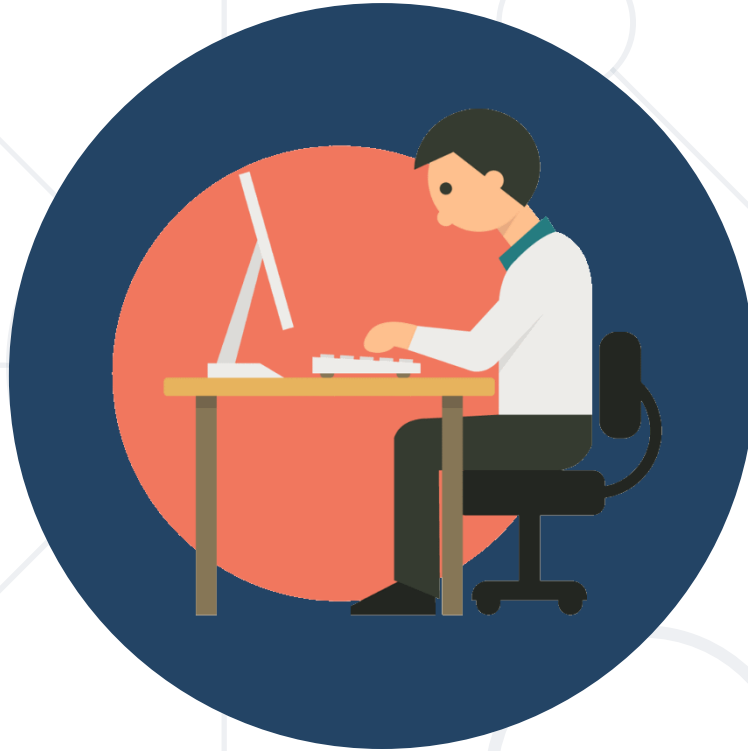
- **Apache Airflow**
 - Workflow Orchestration, Scheduling and Monitoring
 - Example Use Cases – Orchestration ELT Jobs Across Cloud Systems
- **DBT (Data Build Tool)**
 - Focused on Transformations in Modern ELT Pipelines
 - Example Use Cases – Managing Transformations in Snowflake
- Other Tools - **Talend, Apache NIFI, Informatica**
 - Cloud Native – AWS Glue, Goggle Dataflow, Azure Data Factory

1. Retail Use Case: Amazon

- **Scenario:**
- Amazon consolidates sales data from multiple countries to generate revenue reports and optimize inventory.
- Data Sources: Online transactions, regional warehouses, shipping partners.
- **ETL Workflow:**
 - **Extract:** Pull transaction data from multiple country-specific databases.
 - **Transform:**
 - Standardize currency (e.g., convert all sales to USD).
 - Aggregate sales data by region and product category.
 - Clean invalid records (e.g., remove duplicate transactions).
 - **Load:** Store the processed data into a centralized data warehouse (e.g., Snowflake) for reporting.
- **Key Benefits:**
- Enables real-time dashboards for inventory management.
- Provides insights for marketing campaigns based on regional trends.

2. Finance Use Case: Fraud Detection

- **Scenario:**
- A bank uses ETL to clean and load customer transaction data into a centralized database to detect fraudulent activity.
- Data Sources: Multiple banking systems (e.g., ATMs, credit card transactions, mobile apps).
- **ETL Workflow:**
 - **Extract:** Retrieve raw transaction logs from various systems (e.g., ATM transactions, credit card purchases).
 - **Transform:**
 - Normalize transaction timestamps across different time zones.
 - Flag suspicious transactions (e.g., high-value transfers or location mismatches).
 - Enrich the data with customer profiles for context.
 - **Load:** Save the cleaned and enriched data into a fraud detection platform or data warehouse.
- **Key Benefits:**
- Allows near real-time fraud detection using machine learning.
- Reduces false positives by combining transaction and customer data.



Practice

Exercise: Introduction to ETL Processes

- Understand the core concepts of **ETL** and **ELT**
- Learn the **differences** between **traditional ETL** and **modern ELT** processes
- Gain practical skills in implementing ETL pipeline with Python and modern tools

